

Lorenzo Zacchini – Curriculum Vitae

Personal Information

Family name, first name: Zacchini, Lorenzo

Nationality: Italian

Languages: Italian (native), English (fluent)

Email: lorenzo.zacchini@aalto.fi

Institutional webpage: <https://math.aalto.fi/en/people/lorenzo.zacchini>

Personal webpage: <https://lorenzozacchini.github.io/>

Current Position

Doctoral Researcher in Mathematics

Aalto University, Finland

2025 – present

Research Interests

My research interests are primarily in harmonic analysis. In particular, I am currently working on fractional integral operators on spaces of homogeneous type and on the properties of their commutators. Over the course of my studies, I have also explored many other topics—mostly within mathematical analysis but not exclusively—so my curiosity for mathematics extends well beyond my main research area.

Education

Master's Degree in Mathematics

Alma Mater Studiorum – University of Bologna, Italy

2024

Final grade: 110/110 *cum laude*

Academic Visitor

Aalto University, Finland

Feb–May 2024

Scholarship for the preparation of the Master's thesis abroad, awarded by Alma Mater Studiorum – University of Bologna

Erasmus+ Exchange Period

University of Helsinki, Finland

Aug–Dec 2023

Bachelor's Degree in Mathematics

Alma Mater Studiorum – University of Bologna, Italy

2022

Final grade: 110/110 *cum laude*

High School Diploma in Classical Studies

Liceo Ginnasio Luigi Galvani, Bologna, Italy

2018

Final grade: 100/100 *cum laude*

Theses

Master's Thesis: *Schatten Classes, Interpolation Spaces, Singular Integral Operators and their Commutators*

Alma Mater Studiorum – University of Bologna, Italy

2024

Supervisor: Loredana Lanzani

Co-supervisor: Tuomas Hytönen

[link]

Bachelor's Thesis: *La proprietà di media delle funzioni armoniche: rigidità e stabilità*

Alma Mater Studiorum – University of Bologna, Italy

2022

Supervisor: Giovanni Cupini

Co-supervisor: Ermanno Lanconelli

[link]

Teaching Experience

Teaching Assistant, MS-E1461 – Hilbert Spaces

Aalto University, Finland

Period I, Academic Year 2025–2026; Period I, Academic Year 2026–2027

Teaching Assistant, MS-E1531 – Differential Geometry

Aalto University, Finland

Period III, Academic Year 2025–2026

Teaching Support (Curricular Internship), 27213 – Mathematical Analysis 2

Alma Mater Studiorum – University of Bologna, Italy

Second Semester, Academic Year 2023–2024

Talks and Seminars

A Dyadic Approach to Schatten Estimates for Commutators of Fractional Integrals

Midterm Review

Aalto University, Finland

September 23, 2026

Dyadic formula for fractional integrals

XLV Convegno Nazionale di Analisi Armonica

University of Bergamo, Italy

May 26, 2026

Dyadic representation for fractional integrals

ASK (Analysis Students Kernel) Seminar

University of Bologna, Italy

March 31, 2026

Dyadic representation for fractional integrals

FiRST Spring Meeting 2026, Vierumäki, Finland

March 5, 2026

Fractional integrals on spaces of homogeneous type

Analysis Seminar

Aalto University, Finland

February 25, 2026

Conferences and Workshops Attended

12th International Conference on Harmonic Analysis and Partial Differential Equations

El Escorial (Madrid), Spain

June 9–13, 2025

Organized by ICMAT (Instituto de Ciencias Matemáticas) in collaboration with Universidad Autónoma de Madrid

SAB Meeting – Finnish Centre of Excellence in Randomness and Structures (FiRST)

University of Helsinki, Finland

March 13–14, 2025

Espoo, October 1, 2026